Section A

1 **(a)** **(i)** force per (unit) mass ……(*ratio idea essential*) B1

(ii) $g = GM / R^2$ C1
$9.81 = (6.67 \times 10^{-11} \times M) / (6.38 \times 10^6)^2$ ……(*all 3 s.f*) M1
$M = 5.99 \times 10^{24}$ kg .... A0 [2]

(b) **(i)** *either* $GM = \omega^2 r^3$ *or* $gR^2 = \omega^2 r^3$ C1
either $6.67 \times 10^{-11} \text{ x } 5.99 \times 10^{24} = \omega^2 \times (2.86 \times 10^7)^3$
or $9.81 \times (6.38 \times 10^6)^2 = \omega^2 \times (2.86 \times 10^7)^3$ C1
$\omega = 1.3 \times 10^{-4}$ rad s^{-1} .... A1 [3]
(*use of r = 2.22 × 10^{7}m scores max 2 marks*)

(ii) period of orbit = $2\pi / \omega$ C1
= 4.8×10^4 s (= 13.4 hours) A1
period for geostationary satellite is 24 hours (= 8.6×10^4 s) A1
so no .... A0 [3]

(c) satellite can then provide cover at Poles .... B1 [1]

[Total: 10]

2 **(a)** sum of kinetic and potential energies of molecules / particles / atoms M1